

Transfer learning and augmentation

Start from features learned on a broad dataset, then adapt them with examples and transformations that preserve your task.

15–20 MINUTES

Prerequisite: convolution

AUGMENTATION DECISIONS

1 • Pretrain

Learn reusable visual features.

2 • Adapt

Fine-tune for a new target task.

3 • Augment

Create valid transformed examples.

4 • Verify

Measure on untouched conditions.

Why start from pretrained weights?

Early and middle visual features—edges, textures, parts, and spatial patterns—can be useful across tasks. **Transfer learning** starts from parameters learned on a source task instead of random parameters. This often helps when the target dataset is small.

In **fine-tuning**, some or all pretrained parameters continue to update on target data. A fixed-feature approach freezes the feature extractor and trains only a new output layer. Fine-tuning is more flexible but can overfit or damage useful features when data is scarce or updates are too aggressive.

Central mental model

Pretraining supplies a useful starting representation. Target data teaches what must change. Held-out evaluation decides whether the adaptation helped.

Augmentation encodes assumptions

Data augmentation transforms a training example while preserving the intended target. A horizontal flip often preserves “red cube.” A color-channel permutation does not, because color defines the class. Strong blur might be valid if the deployed camera produces blur, but it can erase the very shape cues needed for recognition.

An augmentation is therefore a claim: “the desired answer should remain the same under this transformation.” That claim should follow from deployment conditions, not from a generic list of popular options.

Browser exercise: keep, reject, or investigate

Horizontal flip

Same cube class and valid box after coordinates are updated.

Swap red and blue channels

The visible class color changes.

Moderate exposure change

May represent real camera lighting.

Reveal a defensible decision

Domain adaptation begins with evidence

If a model fails in one environment, adding representative data can help—but first localize the failure and define a test. Mix new data with enough older coverage to avoid improving one setting while forgetting another. Preserve an untouched evaluation set and the previous model artifact.

Applied to our cubes

REPOSITORY OBSERVATION The initial run started from COCO-pretrained YOLO weights rather than random initialization. A later documented fine-tuning run added JetRover-room positives and background-only examples.

EVALUATION BOUNDARY The fact that a training command completed or a checkpoint was saved does not prove the robot-room problem was solved. Compare old and new artifacts on representative held-out positives, negatives, and live conditions.

Project anchor: training and artifact notes.

Check your understanding

What makes an augmentation valid?

- ☐ It always makes training faster
- ☐ It creates brighter output images
- ☐ It preserves the intended target

Explain it back

Explain transfer learning and fine-tuning, then evaluate two augmentations for color-defined cube classes. State the assumption each transformation makes.

Primary sources

- PyTorch transfer learning tutorial.
- D2L: Fine-tuning.